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ORIGINAL PAPER

Social Credit and Trade Credit: A Coevolutionary Perspective

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Abstract
We provide insight into how firms' trade credit decisions respond to the coevolution of business ethics practices, technological innovation, and institutional reform for firms located in pilot cities of China's social credit reform. The reform implements an external monitoring mechanism that potentially shifts the business ethics frontier by punishing or rewarding certain (un)ethical credit behaviors. Following the reform, pilot city firms enjoy greater access to trade credit financing. Three plausible channels include reduced default risk, improved information quality, and enhanced legal protection. Suppliers provide more credit to young, small, private, politically unconnected firms, and those with geographically distant suppliers. Finally, we illustrate how increased access to trade credit translates into tangible strategic and performance benefits. Overall, this study underscores the importance of aligning business ethics practices with evolving societal values and technological advancements to facilitate companies' willingness to provide trade credit within emerging markets.

Keywords Social credit reform · Credit information sharing · Trade credit

JEL Classification G30 · G32 · K40

Introduction

This paper examines the impact of China's social credit reform (SCR) on trade credit dynamics, with a particular focus on its implications for informal financing and institutional development. Employing a coevolutionary framework, we explore how ethical business practices, technological innovation, and institutional changes interact to shape the frontier of business ethics (Wang & Tan, 2023). China's social credit reform (SCR) offers a unique context for this analysis, as its social credit system transcends traditional public credit registries[1] by integrating social and behavioral data alongside financial information (Hou et al., 2024). This broader scope, facilitated by big data and cutting-edge technologies such as artificial intelligence, requires a more robust legal and regulatory framework for managing credit information. To foster a culture of compliance and accountability, the SCR implements clearer and more balanced rewards and penalties while utilizing social monitoring mechanisms to identify and report potential compliance issues.[2]

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[1] According to the World Bank (2020), around 88% of countries had credit bureaus by 2019. For more information, please see https://archive.doingbusiness.org/en/reports/global-reports/doing-business-2020.

[2] The Outline of the Social Credit System Construction Plan (2014–2020) highlights how the SCR strengthens enforcement mechanisms and expands the application of incentives and penalties. For example, companies with poor credit scores may encounter a variety of repercussions, including challenges in securing loans, limitations on eligibility for government contracts, heightened regulatory scrutiny through more frequent inspections, restrictions on market access, reputational damage through public shaming initiatives, and other punitive measures designed to enforce standards. Conversely, entities with strong credit scores benefit from positive incentives. Further details of the

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Our research addresses a significant gap in the literature by investigating the influence of the SCR on firms' ethical credit practices. Unlike prior studies that primarily focused on information sharing (e.g., Brown & Zehnder, 2007; Jappelli & Pagano, 2002) or credit bureaus (e.g., Balakrishnan & Ertan, 2021; Hou et al., 2024), we examine the implications of a comprehensive, publicly accessible platform for sharing information, rewards, and sanctions, integrating data from multiple agencies, government authorities, regulators, the financial system, and courts. This investigation is particularly relevant in the context of trade credit, a form of informal financing in which suppliers bear the risk of credit default from customers who agree to deferred payments. Unlike formal financing, trade credit requires no collateral or guarantees, making it highly dependent on borrowers' creditworthiness (Giannetti et al., 2011), trust (Guiso et al., 2004; Xu et al., 2024), bargaining power (Fabbri & Klapper, 2016), and the level of information asymmetry between suppliers and customers (Uzzi & Lancaster, 2003). In China, trade credit plays an even more critical role than in developed markets (Allen et al., 2019), due to the challenges many firms face in accessing formal financial resources (Ge & Qiu, 2007; Kong et al., 2020; Lai et al., 2022; Quan et al., 2024). Historically, trade credit in China has been heavily reliant on personal connections and trust, often rooted in hometown networks (Kong et al., 2020; Wu et al., 2014). By examining the SCR's role in shaping trade credit dynamics, this study offers insights into the evolving landscape of informal financing in China and provides a broader understanding of how institutional innovations can influence ethical credit practices and economic transactions.

The ex-ante effect of the SCR on trade credit remains unclear. On the one hand, the SCR offers several potential benefits that could facilitate trade credit financing. First, by facilitating the sharing of comprehensive credit information among market participants, the SCR has the potential to mitigate information asymmetry between suppliers and customers, similar to traditional credit bureaus. This improved information flow can empower creditors to make more informed lending decisions.[3] Second, the SCR heightens the reputational consequences of contract defaults,[4] incentivizing prompt debt repayment through its reward and punishment mechanisms. By rewarding responsible financial conduct with higher credit ratings and penalizing unethical behavior, the SCR fosters the evolution of ethical credit practices. Lastly, the SCR strengthens the legal framework governing credit markets by explicitly defining penalties for financial misconduct. This enhanced legal enforcement can bolster creditor confidence and potentially increase the availability of trade credit.

On the other hand, the effectiveness of the SCR in influencing trade credit may be limited by several factors. While the system combines public shaming, business restrictions, and legal consequences, its impact may vary across regions due to differences in enforcement capacity, local political influences, and economic conditions. For example, state-owned enterprises may shield themselves from enforcement actions, particularly in environments with weak investor protection (Ke & Zhang, 2021). Additionally, the SCR's effectiveness is constrained by data fragmentation, as credit data are often isolated across multiple departments and organizations,[5] complicating accurate credit assessments. Finally, the informal nature of trade credit transactions poses challenges to the SCR's effectiveness. Enforcing formal contractual agreements or legal obligations within such informal relationships is inherently difficult, limiting the ability of the SCR to enforce its policies fully. Therefore, while the potential benefits of the SCR in facilitating trade credit are clear, its actual impact remains an empirical question, influenced by a complex set of institutional and practical factors.

To empirically examine the impact of SCR on trade credit, we analyze data from publicly listed firms in China from 2011 to 2019, alongside detailed timelines for the implementation of the SCR. Employing a difference-in-differences (DID) model, we find a significant increase in firms' trade credit following the SCR. Specifically, firms situated in the pilot cities experience an increase of 25% in trade credit and this is primarily driven by the demand side, with a substantial rise in short-term accounts payable. These results remain

Footnote 2 continued
reform are discussed in Sect. "The development of the social credit system in China".

[3] Previous research has consistently demonstrated the positive impact of credit information sharing on market efficiency. These benefits include reduced adverse selection (Pagano & Jappelli, 1993), improved borrower discipline (Bennardo et al., 2015; Padilla & Pagano, 2000), enhanced investment efficiency (Hou et al., 2024), and increased credit availability and affordability (e.g., Jappelli & Pagano, 2002; Kallberg & Udell, 2003; Djankov et al., 2007; Aretz et al., 2020). Furthermore, access to comprehensive credit information improves the accuracy of bankruptcy and default predictions (Bennardo et al., 2015; Nakamura & Roszbach, 2018).

[4] Contract defaults can have severe repercussions, including credit score downgrades, direct penalties (e.g., fines, business closures), and indirect consequences such as limited access to government contracts, higher borrowing costs, increased regulatory scrutiny, and elevated operational expenses.

[5] China's credit information landscape has historically suffered from fragmentation, with data on individuals and businesses scattered across numerous government departments (Qin et al., 2022). These siloed data hinder the effective utilization of collective resources. Furthermore, the market's data ecosystem is characterized by data residing within isolated servers of major operators. The closed and centralized nature of these systems effectively "freezes" data, exacerbating fragmentation. These challenges may collectively impede the digital transformation of the social credit system, making it difficult to accurately and comprehensively assess borrowers' creditworthiness.

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robust through parallel trend analysis and a series of robustness tests, including placebo tests with randomly assigned false event times, propensity score matching (PSM), and a Goodman-Bacon decomposition to account for heterogeneity in treatment timing. Notably, the positive impact of social credit on trade credit has grown since the SCR's implementation, indicating a positive feedback loop between suppliers and customers, both of whom coevolve with the reform's institutional changes and shifting business ethics landscape.

We then explore three plausible channels through which the SCR may influence trade credit financing. First, following the argument by Giannetti et al. (2011) that creditworthy customers are more likely to receive trade credit, we find that the likelihood of default decreases for firms in pilot cities after the reform. Second, consistent with previous studies highlighting the role of the information environment in facilitating trade credit (e.g., Kong et al., 2020; Uzzi & Lancaster, 2003), we find that the SCR improves earnings quality, reduces stock price synchronicity, and increases analyst attention for firms located in pilot cities. Third, we provide evidence for the legal protection channel, as the increase in trade credit is more pronounced in regions with weaker legal protection, supporting the idea that the SCR's external monitoring mechanism—through its systems of rewards and punishments—serves to shift the frontier of business ethics.

Next, we explore and rule out alternative channels, such as the influence of buyers' bargaining power and access to bank credit. Our cross-sectional analysis reveals that the SCR is particularly effective for firms facing greater information asymmetry, including private, unconnected, small, and young firms that typically struggle to secure trade credit. Moreover, the reform has a more significant impact on firms with geographically diverse suppliers, indicating that the reform helps to mitigate information asymmetry between firms and suppliers located in different regions. Notably, the effects of the SCR on trade credit are stronger for those firms located in the "model cities," where the reform's enforcement is more rigorous.

Lastly, we offer empirical evidence that the additional trade credit associated with the reform has real effects on firm strategy and performance. Specifically, we show that the reform's expansion of trade credit leads to more R&D and capital investments, better firm performance as measured by changes in ROE and Tobin's Q, and more environmental initiatives, confirming the importance of informal financing in firms' growth and performance in a typical emerging market (Allen et al., 2019; Ge & Qiu, 2007).

This study contributes to several strands of literature. First, existing studies mainly focus on the effectiveness of public credit registries or bureaus on firms' formal financing and find ambiguous effects of credit information sharing on bank lending (e.g., Balakrishnan & Ertan, 2021; Behr & Sonnekalb, 2012; Grajzl & Laptieva, 2016; Kallberg & Udell, 2003).[6] This paper examines how social credit and ethical business practices, evolving with the institutional setting and technological innovation, promote the sharing of credit information and facilitate informal financing. We show that the positive business ethics frontier shift following the reform, as highlighted by the reduced default risk, improved information quality, and enhanced legal protection channels, contrasts with the typically negative dystopian perspective of the reform in the Western media (Ahmed, 2019). This positive aspect of the reform is consistent with Liu and Liu (2024), who show the reform reduces stock price crash risk.

Second, we extend the literature that examines the determinants of trade credit from a government perspective. Prior studies find that firms' access to trade credit is affected by firm characteristics (e.g., Fabbri & Klapper, 2016; Giannetti et al., 2011), financial and product market characteristics (Cull et al., 2008; Fisman & Love, 2003; Fisman & Raturi, 2004; Uzzi & Lancaster, 2003), and institutional factors (Cai et al., 2023; El Ghoul & Zheng, 2016; Guiso et al., 2004; Kong et al., 2020; Wu et al., 2014; Xu et al., 2024). While Gao et al. (2023) discover that public credit registries create a substitution effect between bank loans and trade credit, our study delves into the broader governance implications of the social credit system. Specifically, we investigate its role in encouraging ethical credit behaviors and facilitating the provision of trade credit.

Third, this study enriches the literature that highlights the importance of informal institutions in promoting economic growth in emerging markets (e.g., Allen et al., 2019; Degryse et al., 2016; Ge & Qiu, 2007; Kong et al., 2020; Lai et al., 2022). Our findings suggest that the additional trade credit financing obtained due to the improved credit information environment promotes firms' strategy, value, and environmental performance in the emerging market of China.

The rest of the paper is organized as follows: Sect. "Institutional background and hypothesis development" presents the institutional background and hypothesis development, Sect. "Sample and methodology" introduces our sample and methodology, Sect. "Empirical results and analysis" presents the empirical results, Sect. "The impact of additional trade credit on firm performance" reports the economic consequence, and Sect. "Conclusion" concludes.

[6] For example, Behr & Sonnekalb (2012) find that the introduction of a public credit registry does not affect access or cost of credit, but improves loan performance. Grajzl & Laptieva (2016) show that information sharing among banks does not affect the volume of bank lending, but sharing among private credit bureaus is associated with higher amount of bank lending. Balakrishnan & Ertan (2021) suggest the establishment of public credit registries, which aims at collecting borrower information significantly improved banks' assessment of loss events.

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Institutional Background and Hypothesis Development

The Development of the Social Credit System in China

The social credit system implemented in China is a government-led initiative aimed at enhancing trust and information exchange among economic participants. In March 2006, the PBC established the *PBC Credit Center* intending to develop a credit system for both individuals and enterprises. The "Outline of the Social Credit System Construction Plan (2014–2020)," published by the *National People's Congress* in 2014, acknowledged the progress made in the social credit system. However, certain challenges remained, such as incomplete credit records for businesses, insufficient incentives for honesty, and inadequate penalties for dishonest or untrustworthy behavior. To address these issues and further refine the social credit system, the NDRC, and *PBC* designated 11 cities as the initial group of pilot cities in August 2015, followed by an additional 32 cities in April 2016 for the implementation of the SCR. The full list of pilot cities is shown in Appendix 1.[7]

As of January 2018, 12 cities were designated as "model cities" for the SCR. The model cities have attached great importance to and carefully organized the creation efforts, effectively advancing the various tasks of building a social credit system as outlined by the national agenda. Since 2016, these cities have consistently ranked among the top nationwide in third-party urban credit status monitoring and have explored and developed a series of replicable and scalable practices. These efforts have played a demonstrative and guiding role in promoting the construction of the urban credit system across the country. In August 2019, the NDRC and PBC jointly announced the second batch of "model cities" for similar reasons, which include 16 different cities and districts.[8]

In pilot cities, a standardized social credit code is implemented to track trust and default, supported by a national platform for sharing credit information and the *Credit China* website. This facilitates data exchange between enterprises and financial institutions, enabling trustworthiness assessments across both financial and non-financial sectors. Geographically distant suppliers can access this information. The SCR builds on existing practices of rewarding trustworthiness and penalizing poor credit but expands and institutionalizes these practices across sectors and regions via a centralized platform.[9] This enhances coordination and transparency, providing a scalable, consistent approach to incentivizing trust and deterring dishonesty. For instance, Yiwu, a global commodity hub, uses market-based incentives, integrating social credit systems with traditional credit reporting to make credit scores more accessible. The city also employs white and blacklists to penalize businesses for significant defaults or contract violations. Wenzhou, known for its entrepreneurial culture, incorporates gamification and public shaming into its credit model, prioritizing financial support for trustworthy businesses and offering incentives such as loans and favorable tax rates. Appendix 2 illustrates two examples from local Credit China websites, featuring data on firms categorized as whitelisted and blacklisted.

Overall, China's social credit system provides critical digital, data-driven infrastructure that enables SCR initiatives. It is a large-scale data collection project with two main goals: (i) To use big data to assess companies' credit information and create the public credit profiles of each business. Companies that score well in the assessment receive fewer random inspections and more favorable loan terms from state-owned banks, and vice versa; (ii) To promote market efficiency and social responsibility through public credit ratings. The first publicly available social credit scores from Zhejiang Province show that neither better-governed nor more profitable firms received higher overall scores.[10] However, highly leveraged firms, which are more likely to default on their loans, are associated with lower total scores and will be subjected to scrutiny. As a whole, the reform facilitates credit information sharing, raises the consequences of default, and improves the social credit landscape in pilot cities.

[7] As noted in Appendix 1, for the cities of Beijing and Shanghai, only the Haidian District of Beijing and the Pudong and Jiading Districts of Shanghai were included in the reform.

[8] These cities were selected based on third-party evaluations by China Reform Daily, Renmin University, Central University of Finance and Economics, and Xi'an Jiaotong University. The first batch included Hangzhou, Nanjing, Xiamen, Chengdu, Suzhou, Suqian, Huizhou, Wenzhou, Weihai, Weifang, Yiwu, and Rongcheng. The second batch of model cities included 16 different cities and districts: Qingdao, Wuhan, Anshan, Pudong, Jiading, Wuxi, Hefei, Huaibei, Wuhu, Anqing, Fuzhou, Putian, Zhengzhou, Yichang, Xianning, and Luzhou.

[9] Designed to improve legal and regulatory enforcement, China's SCR replaced a fragmented system of inconsistent blacklists and redlists with a unified approach. The SCR strengthens rewards and punishments, embodying the principle of "一处失信, 处处受限 " (yī chǔ shī xìn, chù chù shòu xiàn) – "breaking promises or trust in one area results in restrictions everywhere" (Trauth-Goik & Liu, 2023). The Unified Rewards and Punishments mechanism now subjects blacklisted entities to multi-agency sanctions, while redlisted firms receive broader benefits (Schaefer, 2020). Enhanced tracking across agencies and a consolidated public database deter undesirable behavior by increasing transparency and accountability (Drinhausen & Brussee, 2021; Trauth-Goik & Liu, 2023). These changes contribute to a more effective system of rewards and punishments, supported by anecdotal evidence.

[10] For more details, please refer to https://research.hktdc.com/en/article/MTI3NTMxOTEwNg.

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The Coevolutionary Perspective of Ethical Behavior

The conventional approach to business ethics tends to regard ethical behavior as a static set of principles applied irrespective of the external context. However, a recent conceptual coevolution model proposed by Wang and Tan (2023) highlights the interconnectedness of business ethics practices, technological innovation, institutional transition, and disruptive events. Lewin and Volberda (1999) elucidate different properties of coevolution, including *multi-levelness,* wherein micro-coevolution occurs within firms, and macro-coevolution between firms.

At the micro-level, individual companies adapt their credit behaviors in response to internal ethical considerations, as well as the evolving regulatory and technological landscape that shapes credit practices. Meanwhile, at the macro-level, firms collectively influence the broader credit ecosystem through mechanisms such as positive feedback, where ethical practices by one firm serve as an incentive for others to follow suit, and competitive pressures, wherein firms adjust their behavior in line with prevailing practices within the industry. The coevolutionary perspective underscores how business practices and ethical standards evolve alongside other influences (i.e., technological advancements, legal frameworks, and regulatory bodies), and how companies' strategies and approaches adapt to evolving institutional environments (Lewin et al., 1999).

Institutions, defined as "the rules of the game in a society" (North, 1990), play a crucial role in structuring interactions within that society. For example, companies employ different strategic responses to navigate institutional pressures (Oliver, 1991). The three pillars of institutions, regulative (legal and regulatory systems), normative (social norms), and cognitive (shared values and beliefs), collectively define the legitimacy of a firm's strategy concerning adherence to formal laws and regulations, as well as informal norms, values, and culture (Wang & Tan, 2023). Advancements in technology and evolving societal expectations can prompt the establishment of new benchmarks for ethical business conduct. Businesses may adjust their practices to conform to these emerging ethical standards, gradually making them the norm and elevating the overall expectation of ethical behavior.

Firms located in pilot cities may evolve their ethical credit behaviors in response to the external monitoring mechanism. The public-facing SCR acts like the sun, casting light on pilot city firms' credit behaviors. As such, to ensure the 'sunlight' does not reveal poor ethical credit behaviors, firms may engage in micro-coevolutionary responses, actively enhancing their ethical credit behaviors (Brown & Zehnder, 2007), minimizing the withholding of bad news hoarding, and improving their information quality (Liu & Liu, 2024). From a macro-coevolutionary perspective, suppliers are likely to coevolve their credit decisions for customers located within pilot cities. The SCR's 'sunlight' may alleviate information asymmetry and enable suppliers to make well-informed credit decisions (Kallberg & Udell, 2003). The positive feedback loop suggests that suppliers and customers may gradually learn and adapt to each other's credit decisions and ethical credit practices over time.

Hypothesis Development

We propose that the SCR, powered by big data and technological advancements, can facilitate access to trade credit for pilot firms through three key channels.[11] First, social credit can exert a disciplining effect on borrowers by escalating the consequences of unethical credit actions. For example, Dierkes et al. (2013) show that sharing business credit information improves the ex-ante accuracy of default predictions and significantly reduces the ex-post default rates for private firms. As outlined in Sect. "The development of the social credit system in China," China's social credit system and associated credit ratings reward certain good behaviors, while punishing poor credit behavior and dishonest acts. These measures increase the cost of dishonesty and corporate defaults (Yin et al., 2022) and reduce the excess leverage (Sun et al., 2023) as well as risk-taking (Tao et al., 2024). Consequently, firms are more likely to carry out their responsibilities by fulfilling business contracts and reducing opportunistic behavior (Uzzi, 1997). Therefore, we would expect that firms in a good credit-sharing environment are less likely to default, resulting in improved access to trade credit financing.

Second, social credit helps to alleviate uncertainties and reduces credit risk in business transactions (e.g., Jappelli & Pagano, 2002). Previous research indicates that the social credit system discourages financial statement manipulation (Chen et al., 2018), promotes voluntary disclosure (Guan et al., 2020), reduces information asymmetry in corporations (Hilary & Huang, 2023; Liu & Liu, 2024; Yin et al., 2022), and mitigates internal agency issues by facilitating greater corporate risk-taking (Tao et al., 2024). Consequently, a favorable credit environment incentivizes firms to disclose information more transparently, thereby enhancing their ability to access trade credit.

Third, the SCR might facilitate trade credit by enhancing legal protection. Previous research has demonstrated that a robust legal system and adherence to the rule of law are

[11] There is a competing information overload or fatigue argument, where a good credit environment could generate large volumes of information on borrowers, thereby making it difficult for suppliers to accurately access creditworthiness (Padilla & Pagano, 2000). However, on balance we argue the reform should lead to better trade credit decision-making as the pertinent information is often summarized for users. For example, the city of Yiwu's white and black lists are simply summaries of many data points on borrowers' behavior.

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crucial for securing external financing (e.g., La Porta et al., 1998). Firms operating in environments with limited financial market development or weak legal frameworks tend to rely more on informal financing arrangements (Allen et al., 2005, 2019). By imposing punishments and business restrictions, the social credit system could increase trust, as well as the ability of market institutions, to protect the interests of suppliers and investors, thus promoting a robust legal environment (Karlan, 2005). This, in turn, could facilitate firms' access to trade credit. Based on this discussion, we hypothesize as follows (in alternative form):

H1. *Social credit is positively associated with firms' access to trade credit financing.*

However, the SCR may have a negligible impact on suppliers' propensity to extend trade credit to firms located in pilot cities. Trade credit arrangements are frequently informal, rendering it challenging to enforce formal contracts or legal obligations within these business relationships. This can impede the social credit's capacity to monitor and ensure timely payments, particularly in instances where no formal contract or agreement exists. Moreover, the efficacy of local enforcement mechanisms and the quality of data accessible for firms situated in pilot cities may restrict social credit's ability to discipline these firms or furnish credible creditworthiness information to suppliers. These discussions imply that whether social credit improves the trade credit of pilot firms is an empirical question.

Sample and Methodology

Data

The firm-level financial data of listed companies are collected from the *China Stock Market & Accounting Research* (*CSMAR*) database. We exclude financial and special treatment firms, observations with missing data, and firms that were listed after 2016, or delisted before 2015. The regional-level data are obtained from a survey conducted by the "China Enterprise Survey System." All continuous variables are winsorized at the 1% level. Our final sample includes 21,262 firm-year observations of 2731 unique firms from 2011 to 2019. The distribution of pilot cities and companies is presented in Appendix 1.

Identification Strategy

Following Liu and Liu (2024) and Xu et al. (2022), we construct the following DID model to establish the causal effect between social credit and firm-level trade credit.

$$TC_{i,t+1} = \alpha_0 + \beta_1 Reform_{i,t} + \gamma Z_{i,t} + \delta_i + \eta_s + \varepsilon_{i,t}\#, \quad (1)$$

where $TC_{i,t+1}$ denotes trade credit measured as the net amount of accounts payable and accounts receivable scaled by total assets for firm i in year $t + 1$. The key variable of interest $Reform_{i,t}$ is a dummy variable that equals 1 for companies located in the pilot cities and for years during and post the event year, and 0 otherwise. The vector $Z_{i,t}$ denotes a set of control variables which are defined as follows: (i) financial leverage (*Lev*), measured as the ratio of total debt to total assets; (ii) size (*Size*), measured as the natural logarithm of total assets; (iii) cash holding (*Cash*), measured as the ratio of cash divided by total assets; (iv) return on assets (*ROA*), measured as the ratio of net profit to total assets; (v) tangible assets (*PPE*), measured as the ratio of the value of the net property, plants, and equipment divided by the value of total assets; (vi) sales growth (*Growth*), measured as the growth rate of sales; (vii) state-owned enterprises (*SOE*), a dummy variable that equals 1 if the firm is a state-owned enterprise, and 0 otherwise; (viii) firm age (*Age*), measured as the natural logarithm of one plus the number of years since the firm was established; (ix) R&D expense (*R&D*), measured as the ratio of research and development expenses to sales; (x) board size (*Board*), measured as the natural logarithm of the total number of directors; (xi) cash flow (*CFO*), measured as the ratio of cash flow from operating activities divided by total assets; and (xii) independent directors (*Indep*), measured as the ratio of the number of independent directors to the total number of directors. We also include firm (δ) and year (η) fixed effects in our specifications. The coefficient β_1 estimates the effect of the social credit reform on trade credit.

Descriptive Analysis

Table 1 presents the summary statistics. As shown in Panel A, the mean value of the dependent variable *TC* is − 2.4% meaning that on average Chinese firms extend more trade credit (accounts receivable) to their customers than they receive (accounts payable) from suppliers. Accounts payable (*AP*) is 9.2% and accounts receivable (*AR*) is 11.7% of total assets on average. The magnitude of trade credit is similar to that of Wu et al. (2014) and Kong et al. (2020). The statistics of our control variables are also comparable to existing Chinese studies (e.g., Kong et al., 2020). Correlations among our variables are shown in Panel B and are generally below 0.60, suggesting that multicollinearity is less likely to be a major concern.

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Table 1 Summary statistics

Variable	N	Mean	SD	P25	P50	P75
Panel A. Summary statistics						
TC	21,262	− 0.024	0.093	− 0.076	− 0.013	0.033
AP	21,262	0.092	0.069	0.041	0.075	0.124
AR	21,262	0.117	0.103	0.032	0.093	0.172
Reform	21,262	0.224	0.417	0.000	0.000	1.000
Lev	21,262	0.437	0.210	0.268	0.430	0.595
Size	21,262	22.236	1.280	21.327	22.071	22.967
Cash	21,262	0.177	0.126	0.089	0.142	0.228
ROA	21,262	0.033	0.063	0.012	0.033	0.061
PPE	21,262	0.217	0.164	0.089	0.183	0.311
Growth	21,262	0.177	0.446	− 0.026	0.102	0.262
SOE	21,262	0.392	0.488	0.000	0.000	1.000
Age	21,262	2.238	0.731	1.773	2.387	2.886
R&D	21,262	0.016	0.018	0.001	0.013	0.025
Board	21,262	2.137	0.198	1.946	2.197	2.197
CFO	21,262	0.042	0.070	0.004	0.041	0.082
Indep	21,262	0.380	0.072	0.333	0.364	0.429

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)
Panel B. Correlation matrix													
TC	1												
Lev	**0.279**	1											
Size	**0.304**	**0.502**	1										
Cash	**− 0.046**	**− 0.365**	**− 0.229**	1									
ROA	**− 0.059**	**− 0.350**	**0.031**	**0.231**	1								
PPE	**0.205**	**0.072**	**0.082**	**− 0.310**	**− 0.079**	1							
Growth	**− 0.055**	**0.015**	**0.041**	**0.014**	**0.200**	**− 0.091**	1						
SOE	**0.306**	**0.286**	**0.348**	**− 0.098**	**− 0.052**	**0.200**	**− 0.083**	1					
Age	**0.305**	**0.378**	**0.366**	**− 0.265**	**− 0.156**	**0.087**	**− 0.057**	**0.439**	1				
R&D	**− 0.272**	**− 0.257**	**− 0.200**	**0.165**	**0.109**	**− 0.140**	0.011	**− 0.214**	**− 0.250**	1			
Board	**0.124**	**0.142**	**0.251**	**− 0.042**	**0.040**	**0.151**	**− 0.024**	**0.270**	**0.118**	**− 0.095**	1		
CFO	**0.145**	**− 0.157**	**0.065**	**0.129**	**0.346**	**0.255**	− 0.002	**0.018**	0.009	**0.089**	**0.048**	1	
Indep	**− 0.056**	**− 0.059**	**− 0.046**	**0.013**	**0.034**	**− 0.046**	0.004	**− 0.138**	**− 0.107**	**0.073**	**− 0.347**	0.008	1

This table presents summary statistics for the main variables
Panel A presents the basic statistics of the major variables
Panel B presents the Pearson correlation matrix for our main variables, where bold font indicates the variables are significantly correlated at the 10% level or above
All variables are defined in Appendix 3

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Empirical Results and Analysis

The Effect of the SCR on Firms' Access to Trade Credit

We first examine whether the SCR influences firms' access to trade credit. The baseline results are presented in Table 2. Column (1) includes only the independent variable *Reform* and fixed effects, whereas column (2) adds the control variables. The results show that the SCR significantly increases firms' access to trade credit, as the estimated coefficient of *Reform* is 0.007 in column (1) and changes to 0.006 in column (2) with the addition of control variables. In particular, the reform led to a 0.6% increase in the trade credit ratio (equivalent to about 27 million RMB in trade credit),[12] as shown in column (2). Therefore, the baseline results support *H1*.

The coefficients of the control variables are consistent with our expectations. For example, the coefficients of financial leverage (*Lev*), cash ratio (*Cash*), and operating cash flows (*CFO*) are positive and significant, consistent with Wu et al. (2014) and Kong et al. (2020) that firms with higher leverage and operating cash receive more trade credit. The coefficients of *ROA* and *Growth* are negative and significant, reflecting that firms with lower profitability and sales growth rely more on informal trade credit financing. Firm age (*Age*) coefficients are also positive, indicating that older firms enjoy greater access to trade credit financing.

The Demand and Supply Side of Trade Credit

We further examine the impact of the SCR on the demand side (payables) and supply side (receivables) of trade credit. Following Wu et al. (2014), we separate payables/receivables into within 1 year, and more than 1 year to maturity.[13] Therefore, we include total payables/total assets (AP), payables within 1 year/total assets (AP $\leq$ 1 year), and payables of more than 1 year (AP > 1 year) as three separate dependent variables and similarly repeat for receivables.

Table 3 presents the results. The coefficient on the *Reform* variable is positive and statistically significant in columns (1) and (3), where the dependent variable represents either AP or AP $\leq$ 1. This finding suggests a more pronounced impact of the SCR on the demand side of trade credit, incentivizing suppliers to offer shorter-term trade credit to firms within pilot cities. The shorter-term credit is less likely

Table 2 Baseline regressions

Dependent variable =	TC_{t+1} (1)	TC_{t+1} (2)
Reform	0.007***	0.006***
	(3.334)	(2.838)
Lev		0.049***
		(6.714)
Size		0.001
		(0.323)
Cash		0.031***
		(4.925)
ROA		− 0.058***
		(− 4.899)
PPE		0.017*
		(1.803)
Growth		− 0.002**
		(− 2.160)
SOE		0.008
		(1.267)
Age		0.010***
		(3.140)
R&D		− 0.114*
		(− 1.722)
Board		− 0.002
		(− 0.464)
CFO		0.081***
		(9.499)
Indep		0.003
		(0.573)
Constant	− 0.026***	− 0.090**
	(− 52.526)	(− 2.019)
Firm	Yes	Yes
Year	Yes	Yes
Adj. R^2	0.765	0.773
N	21,262	21,262

This table represents the result of regressions that examine the effect of the SCR on trade credit
The dependent variable TC is the difference between accounts payable and receivable divided by total assets
Reform is a dummy variable that equals 1 if firms are registered at pilot cities during and after the event year, and 0 otherwise
All variables are defined in Appendix 3
T-statistics are based on standard errors adjusted for heteroscedasticity and clustered at the firm level
*, **, and *** denote significance at the 10%, 5% and 1% levels, respectively

[12] As the mean value of total assets (*Size*) is 4539 million RMB (= $e^{22.236}$), the reform leads to 27 million RMB more trade credit.

[13] We thank an anonymous reviewer for highlighting the potential rationale for this decomposition. This classification, aligned with common industry practices and prior research (e.g., Wu et al., 2014), recognizes that trade credit transactions are typically settled within shorter timeframes. Longer-term trade credit may represent older receivables or indicate potential difficulties in repayment.

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Table 3 The SCR and demand and supply side of trade credit

	The demand side of trade credit (AP)			The supply side of trade credit (AR)		
	AP	AP (> 1 year)	AP (≤ 1 year)	AR	AR (> 1 year)	AR (≤ 1 year)
	(1)	(2)	(3)	(4)	(5)	(6)
Reform	0.005***	0.000	0.005**	− 0.002	− 0.000	− 0.001
	(2.628)	(0.033)	(2.192)	(− 0.961)	(− 0.070)	(− 0.746)
Lev	0.058***	0.059***	− 0.002	0.001	0.042***	− 0.041***
	(10.407)	(8.396)	(− 0.184)	(0.197)	(5.481)	(− 6.188)
Size	− 0.000	− 0.005***	0.005**	− 0.001	− 0.016***	0.015***
	(− 0.293)	(− 2.702)	(2.128)	(− 0.693)	(− 7.620)	(7.821)
Cash	− 0.023***	− 0.024***	0.001	− 0.040***	− 0.093***	0.053***
	(− 4.531)	(− 3.327)	(0.156)	(− 7.776)	(− 13.179)	(7.918)
ROA	− 0.025***	0.013	− 0.038***	0.026***	0.106***	− 0.079***
	(− 2.831)	(1.310)	(− 3.094)	(2.947)	(8.328)	(− 6.541)
PPE	− 0.032***	− 0.022**	− 0.010	− 0.028***	− 0.082***	0.053***
	(− 4.536)	(− 2.530)	(− 1.014)	(-3.895)	(− 9.065)	(6.054)
Growth	0.002**	0.007***	− 0.005***	0.003***	0.009***	− 0.006***
	(2.427)	(6.149)	(− 3.570)	(4.045)	(8.101)	(− 4.739)
SOE	0.007	− 0.002	0.009	− 0.002	− 0.002	− 0.000
	(1.480)	(− 0.260)	(1.231)	(− 0.390)	(− 0.282)	(− 0.039)
Age	− 0.000	0.005	− 0.005	− 0.007***	− 0.014***	0.007**
	(− 0.007)	(1.238)	(− 1.134)	(− 2.762)	(− 4.247)	(2.382)
R&D	0.154***	0.100	0.054	0.224***	0.499***	− 0.274***
	(3.214)	(1.382)	(0.667)	(3.973)	(6.577)	(− 3.923)
Board	0.005	0.002	0.003	0.002	− 0.004	0.006
	(1.139)	(0.371)	(0.439)	(0.517)	(− 0.772)	(1.211)
CFO	0.019***	0.050***	− 0.031***	− 0.042***	− 0.117***	0.075***
	(2.695)	(5.661)	(− 2.838)	(− 6.811)	(− 12.174)	(7.691)
Indep	− 0.007	0.010	− 0.017**	− 0.007	− 0.003	− 0.004
	(− 1.595)	(1.510)	(− 2.296)	(− 1.439)	(− 0.408)	(− 0.629)
Constant	0.076**	0.131***	− 0.055	0.158***	0.515***	− 0.356***
	(2.271)	(2.918)	(− 1.083)	(4.418)	(11.169)	(− 8.435)
Firm	Yes	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R^2	0.770	0.591	0.570	0.838	0.727	0.519
N	21,262	21,262	21,262	21,262	21,262	21,262

This table reports the regression results of the reform on the demand and supply side of trade credit for pilot firms
The dependent variable in columns (1)–(3) is the proportion of accounts payable divided by total assets, and the dependent variable in columns (4)–(6) is the proportion of accounts receivable divided by total assets
Other variables are defined in Appendix 3
T-statistics are based on standard errors adjusted for heteroscedasticity and clustered at the firm level
*, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively

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to represent aged receivables, potentially signaling reduced repayment difficulties. Several factors may contribute to this increased demand for short-term trade credit. Firstly, the SCR likely enhances information sharing and potentially facilitates social monitoring among entities within pilot cities. This improved information flow mitigates information asymmetry, enabling suppliers to better assess the creditworthiness of potential customers. Consequently, suppliers may feel more confident in extending shorter-term credit terms. Alternatively, the SCR may incentivize local firms to prioritize timely repayment to maintain a positive credit score.

Conversely, as observed in columns (4)–(6) of Table 3, the impact of the SCR on the supply of trade credit (i.e., accounts payable extended by firms to their customers) is relatively limited. This discrepancy may stem from the SCR's primary focus on a firm's internal financial health and regulatory compliance rather than its creditworthiness as a lender. Furthermore, the system may have limited access to data on the creditworthiness of a pilot firm's customers, particularly those located outside of pilot cities. This data limitation hinders the accurate assessment of credit risk associated with extending credit to these entities.

Coevolving with Suppliers: A Positive Feedback Loop

Customers and suppliers may coevolve in response to the SCR, which suggests a feedback loop in trade relationships. More precisely, as firms evolve their ethical credit behaviors in response to the reform's 'sunlight' on their activities, suppliers are likely to observe, learn, and adapt over time, creating a positive feedback loop (Lewin & Volberda, 1999). If a positive feedback loop between suppliers and customers exists, then we would expect pilot city firms to enjoy more trade credit financing as time passes.

To test this, we assess the dynamic effects of the SCR. Specifically, we replace the variable *Reform* in Eq. (1) with a series of dummy variables that trace out the yearly effect of the reform. The dynamic regression equation is presented as follows:

$$\begin{aligned} TC_{i,t+1} = {} & \alpha_0 + \beta_1 Before3 + \beta_2 Before2 \\ & + \beta_3 Current + \beta_4 After1 \\ & + \beta_5 After2 + \beta_6 After3 + \gamma Z_{i,t} + \delta_i + \eta_s + \varepsilon_{i,t} \end{aligned}, \quad (2)$$

where *Before3* and *Before2* are dummy variables indicating the three years and two years before the event year (*Current*), and *After1, After2, and After3* for the one-, two-, and three-years after the event year, respectively. The year before the event year is set as the baseline group.

The results are presented in column (1) of Table 4. The coefficients of *Before3* and *Before2* are statistically insignificant, while the coefficients of *Current*, *After1*, *After2*, and *After3* are significantly positive, suggesting that the increase in trade credit happens only after the reform. In addition, the coefficients for *Current*, *After1*, *After2*, and *After3* increase over time, implying that access to trade credit financing continues to increase for firms located in pilot cities over time. This confirms the positive feedback loop exists between suppliers and customers who learn from and are influenced by each other's behaviors.[14] Figure 1 visualizes the significance of these coefficients. Overall, our analysis suggests that the institutional setting reform is more likely to be an external shock to firms in the pilot cities rather than a perceived response.

Robustness Tests

Placebo Tests

To further verify the authenticity of the SCR, we conduct a placebo test by creating an artificial policy shock. Specifically, we construct a pseudo-policy indicator (*Reform_F*) by advancing the time of policy by two years. The results are presented in column (2) of Table 4. The insignificant coefficient of *Reform_F* indicates that our baseline result is not driven by the potential policy adjustments before the event year.

Also, our identification might stem from confounding factors affecting firms' access to trade credit. To address this concern, we randomly assign pilot cities 1000 times and present the distribution of the falsified coefficients in Fig. 2. The distribution centers around 0, while our true estimate of 0.006 in column (2) of Table 2 falls outside the distribution, suggesting that our baseline results are not biased.

PSM Estimation

We next apply a PSM estimation to address concerns that the fundamental difference between treatment and control firms could potentially confound our results. Specifically, we denote firms located in pilot cities as the treatment group and match the treatment with a control group based on a one-to-one nearest-neighbor matching without replacement. The propensity scores are obtained by running a logit regression on *Reform* with control variables in Eq. (1). Then, we re-run Eq. (1) using the matched sample. Column (3) of Table 4 presents the estimation results.[15] Our coefficient of *Reform* is positive and significant at the 10% level, reinforcing our argument that firms located in pilot cities have better access to trade credit after the reform.

[14] The results presented in this section also validate the parallel trend assumption underlying our DID approach.

[15] After the matching procedure, the differences between the two groups become less significant for the majority of our control variables. The univariate analysis results are presented in Appendix 5.

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Table 4 Robustness tests

	Parallel trend	Placebo Tests	PSM-DID	Goodman-Bacon decomposition
Dependent variable =	TC_{t+1}	TC_{t+1}	TC_{t+1}	TC_{t+1}
	(1)	(2)	(3)	(4)
Reform			0.006*	0.006***
			(1.694)	(2.401)
Reform_F		0.010		
		(1.419)		
Before3	− 0.002			
	(− 1.058)			
Before2	0.000			
	(0.178)			
Current	0.003*			
	(1.712)			
After1	0.004*			
	(1.771)			
After2	0.007***			
	(2.649)			
After3	0.009***			
	(3.067)			
Treated v Never T				0.0064
				[weight = 0.938]
Early T. v Late C				0.0013
				[weight = 0.031]
Late T. v Early C				0.0010
				[weight = 0.031]
Controls	Yes	Yes	Yes	Yes
Firm	Yes	Yes	Yes	–
Year	Yes	Yes	Yes	–
Adj. R^2	0.773	0.628	0.802	–
N	21,262	21,262	6887	–

This table presents the robustness results. Column (1) reports the results for the parallel trend, where *Reform* is replaced with a number of dummy variables that trace out the yearly effects of the SCR
Column (2) reports the placebo test results where we assume the policy shock occurs two years before the actual policy.
Column (3) presents the estimation results using a PSM-matched sample
Column (4) reports the results using Goodman-Bacon decomposition to address the heterogeneity of treatment effects
Controls are the same as those in Table 2, with detailed variable definitions in Appendix 3
T-statistics are based on standard errors adjusted for heteroscedasticity and clustered at the firm level
*, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively

Goodman-Bacon Decomposition

Goodman-Bacon (2021) highlights the potentially misleading nature of the two-way fixed effects (TWFE) DID approach when average treatment effects exhibit variation over time. Specifically, if treatment effects remain consistent over time, TWFE DID yields a variance-weighted average of treatment effects across different groups, with all weights being positive. However, when previously treated units are utilized as controls, negative weights emerge, necessitating the subtraction of changes in their outcomes. This adjustment accounts for potential time-varying treatment effects that may be involved. To address the concern that our results are driven by a misleading staggered DID estimation, we conduct the method suggested by Goodman-Bacon (2021).

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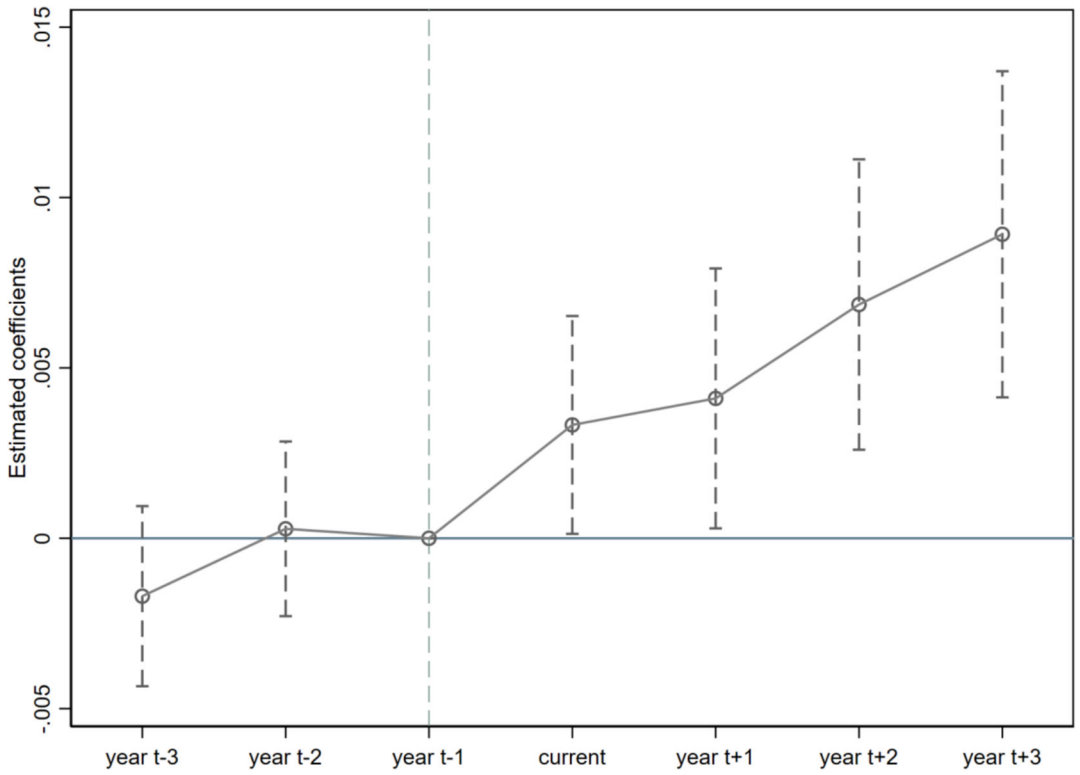


Fig. 1 Dynamic analysis. This figure shows the time-variant roles of the SCR on trade credit. The effect of the reform on trade credit is insignificant before the baseline year, which verifies the existence of a parallel trend between the control and treatment groups. It becomes significantly positive after the baseline year, indicating the reform could enhance trade credit

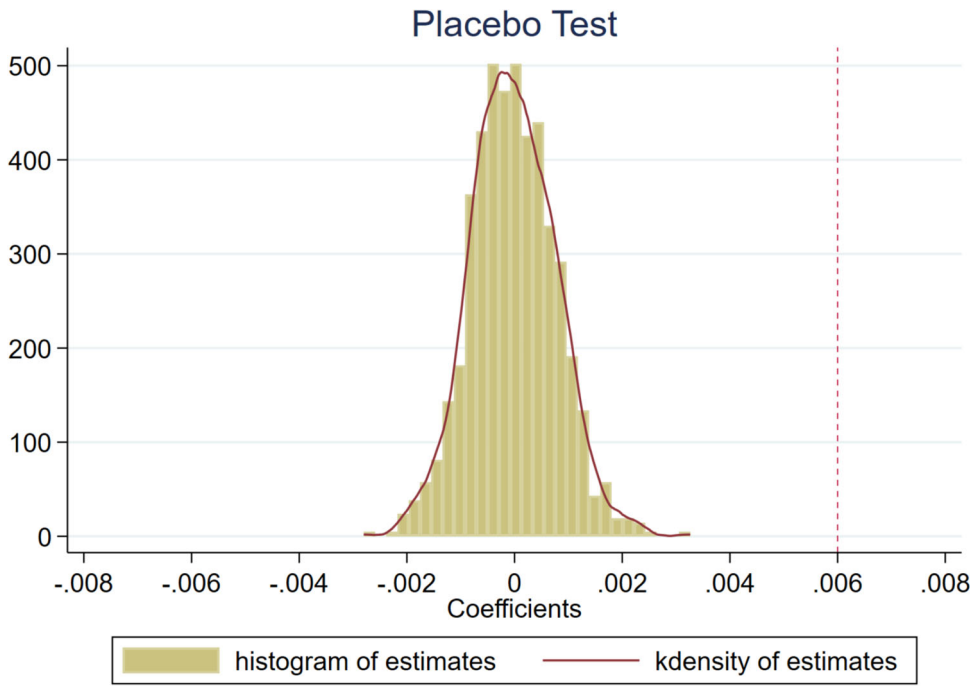


Fig. 2 Placebo tests. In this figure, the Kernel density and 1000 estimates of the *Reform* coefficients are depicted, which were generated by randomly perturbing pilot cities. It can be observed that the distribution of these simulated estimates primarily centers around zero, and notably, the actual estimate (0.006, as indicated in column 2 of Table 2) falls outside the range of the placebo test. These findings indicate that our estimation results are not significantly influenced by any unaccounted variables, thus suggesting minimal bias

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Column (4) of Table 4 reports the Goodman-Bacon decomposition and shows that after considering decomposition effects, our core variable *Reform* remains positive and significant at the 1 percent level. The positive effect is mainly driven by the treatment group (*Treated*) and the control group (*Never T.*), which has the highest weight of 0.938, while the contaminated group weight (*Late T. v Early C.*) is only 0.031. As such, our results are mainly due to differences between the pilot cities and cities that have never been part of the treatment group, confirming the robustness of our baseline findings.

Other Robustness Tests

To further assess the robustness of our findings, we conduct a series of additional analyses. These included ruling out the probability that the selection of pilot cities is based on city-level characteristics, using alternative measures of trade credit and different sample periods, experimenting with different fixed effect combinations and clustering effects, controlling for city-level factors and contemporaneous policies, and addressing potential omitted variable bias. The results of these analyses are detailed in Appendices 6 and 7. In addition, Fig. 3 presents the outcomes of a comprehensive specification curve analysis encompassing 72 models with various specification choices. Collectively, the staggered DID results and the robustness tests strongly suggest a positive and causal relationship between the reform and firms' access to trade credit.

Potential Channels

In this section, we explore potential channels through which the SCR increases the trade credit of pilot firms. We contend that the reform's 'sunlight' reveals, encourages, and rewards ethical credit behaviors such as diminished default risks, and enhanced information quality. It also penalizes unethical conduct by blacklisting firms that breach contracts and impose restrictions on their operations (such as being prohibited from expanding or relocating), thereby safeguarding the interests of investors and suppliers.

Reduced Default Risks

The SCR can have a disciplining effect on market participants by encouraging borrowers to meet their payment obligations promptly, subsequently reducing suppliers' credit risk. When borrowers know that their credit history is being publicly shared, they are more likely to repay their loans on time to maintain a good credit score. Suppliers benefit from social credit reform because it helps them to better assess the creditworthiness and default risk of borrowers (Quan et al., 2024). This allows them to make more informed lending decisions and reduce their risk of loss. Consequently, suppliers are more inclined to offer higher levels of trade credit (Guan et al., 2020) to firms in pilot cities.

To empirically examine the default risk channel, we employ three proxies to measure firm-level default risk: (i) the natural logarithm of managerial default frequency (*Default*)[16]; (ii) negative of distance to default (*DD*) based on Bharath and Shumway (2008); (iii) negative of Z-score (*Z-score*) based on Aretz et al. (2020). Higher values of these measures suggest a greater risk of a company defaulting on its obligations. We first examine whether the reform reduces firm-level default risks by using these three measures as dependent variables. To determine whether pilot city firms evolve their ethical business practices in response to the SCR, we further incorporate interactions between the *Reform* and changes in each default risk measure during the pre- to post-reform period. This approach allows us to assess whether the impact of SCR on trade credit is mediated through lower default risk.

The empirical results are presented in Table 5. The significant and negative coefficients of *Reform* in columns (1), (3), and (5) show that SCR significantly reduces default risk in pilot city firms. Moreover, the negative interaction terms of *Reform* and changes in default risk proxies in columns (2), (4), and (6) suggest that the positive impact of the SCR on trade credit is more pronounced in pilot city firms that actively reduce their default risk. Overall, our results suggest that the SCR has improved the business ethics frontier through the creditworthiness enhancement of pilot city firms, thereby improving their access to trade credit. The positive shift in the business ethics frontier is consistent with pilot cities' improved business credit environment ratings relative to non-pilot cities following the reform as shown in Appendix 4.

Improved Information Quality

We next examine whether the positive impact of social credit on trade credit is due to improved information quality. The reform encourages credit information sharing, and suppliers might have better access to customers' credit history and the financial situation of firms located in the pilot cities. To verify this conjecture, we first examine the impact of the reform on information quality, proxied by the negative level of discretionary accruals (*EQ*), negative stock price synchronicity (*IN*),[17] and analyst attention (*Attention*). Detailed variable explanation is presented in Appendix 3. To further establish

[16] Managerial default occurs when the management fails to act on a given situation such as failure to meet targets or goals, or neglecting legal or ethical responsibilities. The data come from the CSMAR database.

[17] Based on the Modified Jones (1991) model, we first estimate the following equation cross-sectionally for each year and industry to obtain

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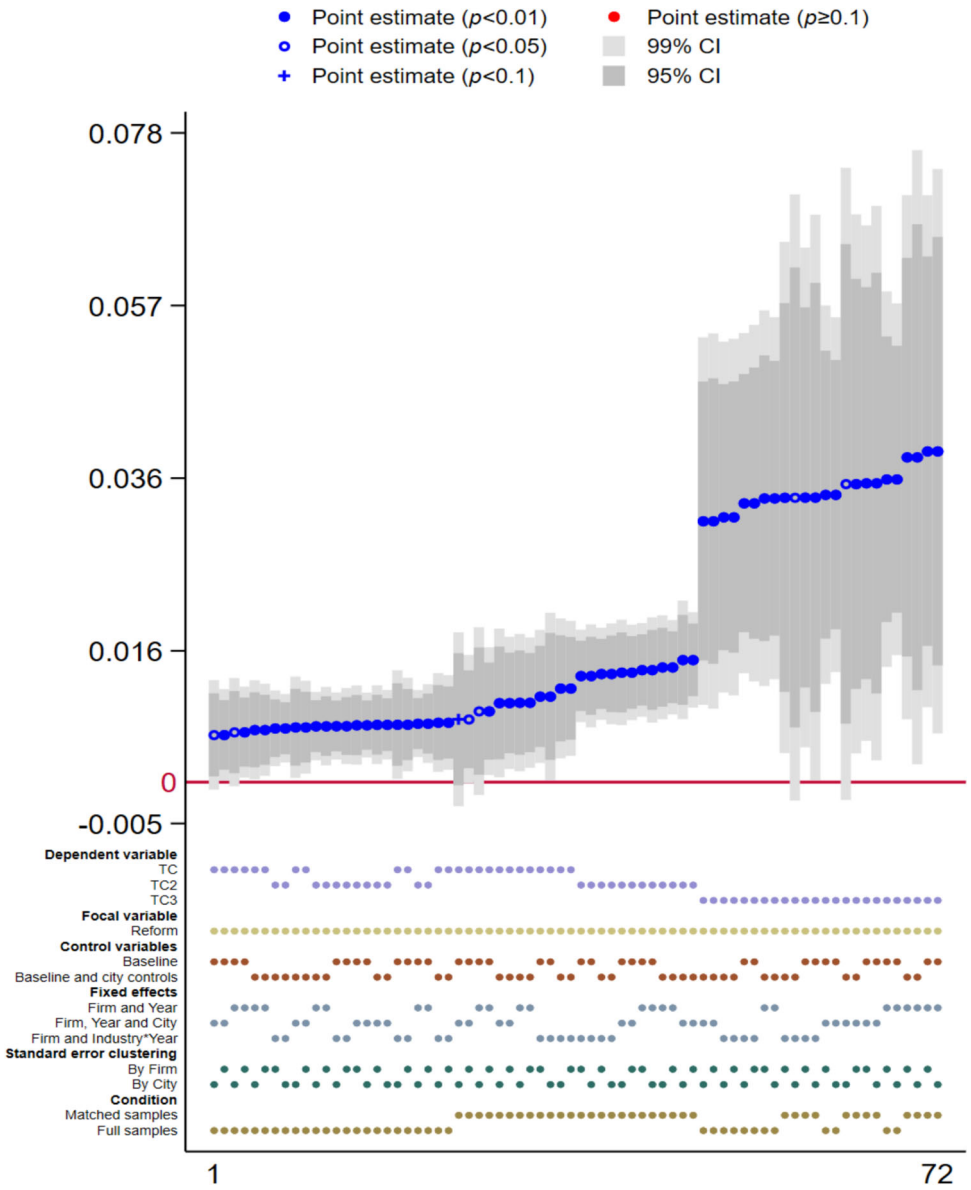


Fig. 3 Specification curve analysis. Displayed in this figure are the outcomes of a specification curve analysis exploring the connection between the reform and trade credit. It showcases the coefficient estimates of *Reform* across 72 model specifications. To simplify, we annotate solely the maximum and minimum coefficient estimates, alongside the zero threshold at both the 95% and 99% confidence intervals. Notably, across all specifications, we observe the coefficient estimates of *Reform* to be statistically significant at the 1% level

the channel, we also incorporate an interaction term between the change in information quality pre- and post-reform in Eq. (1).

The results are reported in Table 6. The coefficients of *Reform* are negative and statistically significant at the 1% level in columns (1), (3), and (5), suggesting that the

Footnote 2 continued
the estimated accruals:

$$ACCRUAL_{it} = \alpha_0 + \alpha_1\left(\frac{1}{ASSETS_{it-1}}\right) + \alpha_2 (\Delta SALES_{it} - \Delta AR_{it}) + \alpha_3 PPE_{it} + \epsilon_{it},$$

where i and t denote firm and year, respectively. *ACCRUAL* is total accruals scaled by lagged total assets ($ASSETS_{it-1}$). $\Delta SALES$ is change in sales scaled by $ASSETS_{it-1}$. ΔAR is change in account receivable scaled by $ASSETS_{it-1}$. PPE_{it} is net property, plant, and equipment scaled by $ASSETS_{it-1}$. ϵ is the error term.

Footnote 17 continued

EQ is the negative of discretionary accruals, calculated as the difference between the actual $ACCRUAL$ and estimated $ACCRUAL$ from the above equation.

Following Jin and Myers (2006), we first apply the below equation on stock returns of a firm:

$$r_{i,d,t} = \alpha + \beta_{1i} r_{m,d-2,t} + \beta_{2i} r_{m,d-1,t} + \beta_{3i} r_{m,d,t} + \beta_{4i} r_{m,d+1,t} + \beta_{5i} r_{m,d+2,t} + \epsilon_{i,d,t}\#,$$

where ri,d,t is the daily return of stock i on day d of year t, and rm,d,t is the value-weighted A-share market return on day d of year t.

To address the bounded nature of R^2 within the range [0, 1], below equation is applied to calculate *SYN*, and *IN* is the negative of *SYN*.

$$SYN_{i,t} = \log\left(\frac{R^2_{i,t}}{1-R^2_{i,t}}\right)\#$$

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Table 5 The default risk channel

Dependent variable =	*Default* (1)	TC_{t+1} (2)	*DD* (3)	TC_{t+1} (4)	*Z-score* (5)	TC_{t+1} (6)
Reform	− 0.017***	0.004**	− 0.145***	0.004**	− 0.371***	0.006***
	(− 7.569)	(2.093)	(− 3.137)	(2.013)	(2.625)	(2.838)
$(Default_{post}—Default_{pre}) \times Reform$		− 0.264***				
		(− 3.855)				
$(DD_{post}—DD_{pre}) \times Reform$				− 0.003***		
				(− 3.020)		
$(Z\text{-}score_{post}—Z\text{-}score_{pre}) \times Reform$						− 0.001**
						(− 2.117)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Firm	Yes	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R^2	0.106	0.774	0.690	0.774	0.673	0.774
N	21,262	21,262	21,015	21,262	21,015	21,262

This table shows the results of the default risk channel, where default risk is proxied by the frequency of managerial default (*Default*), negative of distance to default (*DD*), and negative of Altman's *Z-score*
Higher values of *Default*, *DD,* and *Z-score* signal a greater risk of default
Controls are the same as those in Table 2 and all variable definitions are shown in Appendix 3
T-statistics are based on standard errors adjusted for heteroscedasticity and clustered at the firm level
*, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively

Table 6 The information quality channel

Dependent variable =	*EQ* (1)	TC_{t+1} (2)	*IN* (3)	TC_{t+1} (4)	*Attention* (5)	TC_{t+1} (6)
Reform	0.008***	0.005**	0.120***	0.007***	0.069**	0.006***
	(2.607)	(2.414)	(3.635)	(3.298)	(2.349)	(2.933)
$(EQ_{post}—EQ_{pre}) \times Reform$		0.060***				
		(2.790)				
$(IN_{post}—IN_{pre}) \times Reform$				0.004**		
				(2.127)		
$(Attention_{post}—Attention_{pre}) \times Reform$						0.007***
						(3.262)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Firm	Yes	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R^2	0.195	0.774	0.355	0.774	0.664	0.774
N	20,579	21,262	20,669	21,262	21,262	21,262

This table shows the results of the information quality channel, where information quality is measured by negative discretionary accruals (*EQ*), negative stock price synchronicity (*IN*), and analyst attention (*Attention*), respectively
Higher levels of EQ, IN, and Attention indicate better information quality
Controls are the same as those in Table 2 and all variable definitions are shown in Appendix 3
T-statistics are based on standard errors adjusted for heteroscedasticity and clustered at the firm level
*, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively

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reform improves firms' information quality and availability. In addition, the interaction terms of *Reform* and changes in information quality measures are statistically significant and positive in columns (2), (4), and (6). Overall, our findings confirm the information quality channel and are consistent with Liu and Liu (2024) who show that the reform reduces pilot city firms' crash risk.

Improved Legal Protection

The effectiveness of a legal system significantly impacts a firm's ability to secure external financing (e.g., La Porta et al., 1998). In regions with underdeveloped financial markets or weaker legal frameworks, firms often rely more heavily on informal financing (Allen et al., 2005, 2019). The social credit system not only fosters trust (Yin et al., 2022) but also strengthens the legal environment by imposing direct (fines) and indirect penalties (limited participation in government contracts, loans, or business activities) on firms that breach contracts. This improved legal protection can enhance suppliers' willingness to extend trade credit by safeguarding their interests and fostering trust. Therefore, the positive impact of the social credit system on trade credit may be attributed, in part, to its role in enhancing legal protections. To examine this hypothesis, we follow Wu et al. (2014) by (i) using the marketization index (*Market*); (ii) lawyer-to-population in a prefectural city (*Law*); and (iii) the investor protection index (*IPI*) measured by the fifth component of the China Marketization Index (Yu and Pan, 2008; Pan et al., 2009) as proxies for the formal institutions. We then interact with *Market*, *Law,* and *IPI* with *Reform* in Eq. (1), respectively.

The results are presented in Table 7. The coefficients of *Market* × *Reform Law* × *Reform,* and *IPI* × *Reform* are negative and statistically significant, suggesting that regions with low marketization, poor access to legal representation, and weak investor protection, experience a greater increase in trade credit after the SCR. Therefore, improved legal protection may be a channel that drives our results.

Table 7 The legal protection channel

Dependent variable =	TC_{t+1} (1)	TC_{t+1} (2)	TC_{t+1} (3)
Reform	0.099***	0.009***	0.030***
	(5.332)	(3.408)	(3.744)
Market	− 0.007*		
	(− 1.862)		
Market × *Reform*	− 0.042***		
	(− 5.110)		
Law		− 0.146	
		(− 0.179)	
Law × *Reform*		− 0.797**	
		(− 1.964)	
IPI			− 0.010***
			(− 3.803)
IPI × *Reform*			− 0.014***
			(− 3.437)
Controls	Yes	Yes	Yes
Firm	Yes	Yes	Yes
Year	Yes	Yes	Yes
Adj. R^2	0.775	0.774	0.774
N	21,252	21,262	21,234

This table presents the results of the legal protection channel, where legal protection is measured by the marketization index (Market) in column (1), lawyer-to-population in a prefectural city (Law) in column (2), and the investor protection index (IPI) measured by the fifth component of the China Marketization Index in column (3)
Controls are the same as those in Table 2 and all variable definitions are shown in Appendix 3
T-statistics are based on standard errors adjusted for heteroscedasticity and clustered at the firm level
*, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively

Cross-Sectional Analysis

Certain firm characteristics or business environments have traditionally affected suppliers' willingness to extend trade credit finance. If the reform's 'sunlight' reveals and encourages good ethical credit behaviors, we would expect pilot city firms who previously enjoyed less trade credit financing due to information asymmetry to benefit the most from the reform. In addition, there might be differences in pilot cities when implementing the social credit construction reform. Therefore, in this subsection, we examine whether firm characteristics, the distance between the firm and its suppliers, and variations in the SCR implementation affect the relationship between social credit and trade credit.

The Role of Firm Characteristics

Firm attributes can serve as indicators of information asymmetry and supplier dependence (Petersen & Rajan, 1997). A well-functioning social credit system can enhance information transparency, potentially mitigating default risks. This, in turn, is expected to have a more pronounced effect on trade credit availability for specific firm categories. For example, private firms and those lacking strong political connections often face limitations in accessing formal financial markets (Ge & Qiu, 2007; Liu et al., 2013), and therefore may strive to improve creditworthiness to obtain informal financing. Smaller and younger firms typically possess fewer assets, making it challenging to establish their creditworthiness (Beck & Demirguc-Kunt, 2006). Moreover, young firms, by

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nature, have not had the opportunity to build trust and rapport with suppliers. Therefore, we posit that a social credit system, by improving information quality and reducing perceived default risks, would incentivize suppliers to extend credit to customers they might have previously been hesitant toward, due to concerns about information asymmetry and potential defaults.

To empirically test this hypothesis, we create the following four variables: (i) *SOE*, a dummy variable that indicates whether a firm is state-owned; (ii) *PC*, which takes the value of 1 if the CEO or Chairman of a firm is a former government official, a member of the National People's Congress (NPC), or a member of the Chinese People's Political Consultative Conference (CPPCC), and 0 otherwise (Liu et al., 2013); (iii) *Size*, the natural logarithm of total assets; (iv) *Age*, the natural logarithm of years since a firm's founding. Then, we interact these variables with *Reform* in Eq. (1), respectively.

The results are shown in Panel A of Table 8. The coefficient of the interaction variables *SOE* × *Reform* and *PC* × *Reform* is negative and significant in columns (1) and (2), confirming our expectation that the effect of the reform is more pronounced for non-SOEs and private firms without political connections. In column (3), the negative coefficient of *Size* × *Reform* suggests that small firms located in pilot cities access more trade credit after the reform. Similarly, the negative coefficient of *Age* × *Reform* in column (4) implies that younger firms in pilot cities enjoy greater access to trade credit financing following the reform. Overall, the results suggest that firms that previously struggled to attract trade credit financing, due to higher levels of information asymmetry, enjoy greater access to trade credit after the social credit environment is improved.[18]

The Role of Geographic Distance

The gravity literature has highlighted the significant and growing impact of geographic distance in facilitating information exchange within supply chains (Huang et al., 2021) and corporate financing (El Ghoul et al., 2013). Suppliers encounter additional challenges and costs in obtaining information, such as assessing customer creditworthiness and monitoring, when the buyer and supplier relationship spans a considerable geographical distance, leading to increased non-payment risk (Wilson & Summers, 2002). Additionally, establishing personal connections becomes more challenging for firms located far from their suppliers. The reform should help resolve suppliers' challenges and costs associated with distance. As such, we anticipate the reform will have a greater impact on firms with more geographically dispersed business relationships. We use both a dummy variable indicating whether at least one of the biggest five suppliers is located in a different city as the firm (*Different_city*) and the average distance between a firm with its five biggest suppliers (*Distance*) as proxies for the geographic distance. We expect the reform to have a bigger impact on firms with suppliers located further away.

To empirically test the role of geographic distance on the effects of the reform on trade credit, we interact *Different_city* and *Distance* with *Reform* in Eq. (1), respectively, and present the results in Panel B of Table 8. The coefficients of *Different_city* × *Reform* and *Distance* × *Reform* are positive and statistically significant at the 5% level and 10% level, respectively. The results confirm our conjecture that social credit reform has a greater impact when buyers and suppliers are geographically distant.

The Role of the SCR Implementation Intensity[19]

Despite the NDRC and PBC designating 43 regions across 42 cities for SCS development, implementation varied significantly. As of January 2018, based on third-party evaluations by *China Reform Daily*, *Renmin University*, *Central University of Finance and Economics*, and *Xi'an Jiaotong University*, only 12 cities were deemed 'model cities.' These model cities, all included in the initial two batches of social credit construction, demonstrated greater effort in implementing the system. To analyze whether these model cities have a more pronounced effect on firms' trade credit, we created a dummy variable, *Model city*, equaling one for firms located in designated model cities and zero otherwise. We then interact *Reform* with *Model city* and included this interaction term in our baseline regressions. The results, presented in Panel C of Table 8, reveal a positive and statistically significant coefficient for *Reform* × *Model city*, suggesting that the positive impact of social credit reform on trade credit is more pronounced in 'model cities.'

Ruling Out Alternative Explanations

The Effect of Bargaining Power

In the context of increased information transparency and public scrutiny, listed firms in pilot cities may cultivate a

[18] In untabulated results, we find that corporate governance as measured by the corporate governance index and institutional ownership (etc.), and financial constraints as measured by the cash flow sensitivity and Whited and Wu (WW) index, do not moderate the effects of the reform. As such, firm ownership structure, political connections, size, age, and geographic distance rather than a firm's corporate governance and financial constraints, are likely to affect the relationship between social credit and trade credit. Therefore, the reform enables greater access to a vital source for financing growth opportunities in emerging markets for these types of firms (Allen et al., 2019), which are often underserved by China's formal financing channels (Ge & Qiu, 2007).

[19] We thank an anonymous reviewer for the suggestion.

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Table 8 Cross-sectional analysis

Dependent variable	TC_{t+1} (1)	TC_{t+1} (2)	TC_{t+1} (3)	TC_{t+1} (4)
Panel A. The role of firm characteristics				
Reform	0.007***	0.008***	0.066**	0.025***
	(2.777)	(2.603)	(2.041)	(3.500)
SOE × Reform	− 0.018***			
	(− 4.325)			
SOE	− 0.008**			
	(− 2.208)			
PC × Reform		− 0.009**		
		(− 2.031)		
PC		− 0.005*		
		(− 1.849)		
Size × Reform			− 0.003*	
			(− 1.951)	
Size			− 0.003	
			(− 1.493)	
Age × Reform				− 0.008***
				(− 2.663)
Age				0.002
				(0.576)
Controls	Yes	Yes	Yes	Yes
Firm	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes
Adj. R^2	0.774	0.746	0.774	0.773
N	21,262	12,936	21,262	21,262

Dependent variable	TC_{t+1} (1)	TC_{t+1} (2)
Panel B. The role of geographic distance		
Reform	0.007***	− 0.002
	(3.113)	(− 0.449)
Different_city	− 0.002	
	(− 1.561)	
Different_city × Reform	0.004**	
	(2.037)	
Distance		− 0.000
		(− 0.506)
Distance × Reform		0.001*
		(1.891)
Controls	Yes	Yes
Firm	Yes	Yes
Year	Yes	Yes
Adj. R^2	0.773	0.773
N	21,262	21,262

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Table 8 (continued)

Dependent variable =	TC_{t+1} (1)
Panel C. The role of the SCR implementation intensity	
Reform × *Model city*	0.010**
	(2.494)
Reform	0.004*
	(1.864)
Model city	0.016
	(1.185)
Controls	Yes
Firm	Yes
Year	Yes
Adj. R^2	0.774
N	21,262

This table shows the cross-sectional analysis results
Panel A examines the role of firm characteristics in affecting the relationship between the SCR and trade credit
Firm characteristics include ownership structure (SOE), political connection (PC), firm size (Size), and firm age (Age)
Panel B presents the results examining the effect of geographic distance on the positive relationship between the SCR and trade credit
Geographic distance is represented by Different_city, a dummy that equals 1 if the supplier is located in the same city as a firm, and 0 otherwise; and Distance, the average distance between a firm and its five biggest suppliers
Panel C displays the results examining the effect of SCR implementation intensity on the relationship between the SCR and trade credit
Model city is a dummy variable that equals one for firms that were designated as "model cities" based on third-party evaluations, and zero otherwise
Controls are the same as those in Table 2 and all variable definitions are shown in Appendix 3
T-statistics are based on standard errors adjusted for heteroscedasticity and clustered at the firm level
*, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively

reputation for greater trustworthiness following the reform. This enhanced standing could facilitate smoother business operations, providing a competitive edge in negotiations with suppliers, and being especially advantageous for firms with limited bargaining power before the reform (Fabbri & Klapper, 2016). The positive impact of the reform on trade credit may be due to the increased bargaining power of pilot city firms.[20]

To rule out this explanation, we divide the sample into two distinct groups based on the firm's industry position within the supply chain (upstream or downstream) and their median level of market share, respectively. Firms operating in upstream industries and those with higher market share possess stronger negotiating power when dealing with suppliers. Specifically, we categorize firms engaged in extractive industries (B), metal smelting and equipment manufacturing (C3), electricity, gas, and water production and supply (D), transportation and storage (G), information technology (I), and communication and cultural (R) industries as belonging to the upstream group, while firms in other industries are classified as downstream. Moreover, we measure market share by the proportion of a firm's sales in the industry (Lai et al., 2022). If the buyers' market hypothesis holds, we would expect firms with stronger bargaining power to have greater access to trade credit following the reform.

The results are presented in columns (1)-(4) of Panel A, Table 9. The differences in coefficients are not significant between firms with high and low bargaining power, indicating that the rise in trade credit is unlikely attributable to the affected firms' market power. Thus, our findings do not support the buyers' market hypothesis.

The Effects of Bank Financing on Trade Credit

According to Allen et al. (2019), formal and informal financing can exhibit characteristics of both substitutes and

[20] Our unreported findings suggest that the SCR is associated with an increase in market share for locally listed companies, as measured by their proportion of sales within their respective industries. We thank an anonymous referee for suggesting this important finding.

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Table 9 Ruling out alternative explanations

	Upstream		Market share	
	Yes	No	High	Low
	(1)	(2)	(3)	(4)
Dependent variable =	TC_{t+1}	TC_{t+1}	TC_{t+1}	TC_{t+1}
Panel A. Bargaining power theory				
Reform	0.006**	0.007*	0.008***	0.006*
	(2.302)	(1.724)	(2.903)	(1.822)
Controls	Yes	Yes	Yes	Yes
Firm	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes
Adj R^2	0.778	0.769	0.786	0.775
N	12,575	8687	10,631	10,631
Difference between groups	$X^2 = 0.08$ p = 0.771		$X^2 = 0.22$ p = 0.639	

Dependent variable =	Positive	Negative	Zero
	(1)	(2)	(3)
Panel B. Credit substitution theory			
Reform	0.008***	0.008***	0.003
	(3.058)	(2.595)	(0.886)
Controls	Yes	Yes	Yes
Firm	Yes	Yes	Yes
Year	Yes	Yes	Yes
Adj. R^2	0.779	0.782	0.826
N	10,830	7622	2832

This table reports the results of alternative explanations regarding the increase in trade credit following the SCR
Panel A presents the regression results examining bargaining power explanations for the increase in trade credit after the construction of a treatment city
Upstream takes 1 if firm located in upstream industries for extractive industries (B), metal smelting and equipment manufacturing (C3), electricity, gas, and water production and supply (D), transportation and storage (G), information technology (I), and communication and cultural industries (R), and 0 otherwise. *Market share* is measured by a firm's proportion of sales in the industry
Panel B investigates the impacts of bank loans on the association between the SCR and trade credit
Positive indicates that bank credit and trade credit are complements, *Negative* indicates that they are substitutes, and *Zero* indicates either missing data or no changes in either trade credit or bank credit
All variables are defined in Appendix 3
T-statistics are based on standard errors adjusted for heteroscedasticity and clustered at the firm level
*, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively

complements, leading to an ambiguous relationship between trade credit and bank credit. For instance, Chen et al. (2019) suggest a potential substitution between trade credit and bank credit, indicating that firms may not require extensive trade credit if bank credit is easily accessible. On the other hand, Cook (1999) argues that during financial crises, information asymmetry between banks and firms increases, and trade credit can serve as a means for banks to gather information. Consequently, banks may provide bank credit to companies that have access to greater trade credit. Additionally, Shenoy and Williams (2017) demonstrate that suppliers with better access to bank liquidity are more inclined to offer higher levels of trade credit to customers. Del Gaudio et al. (2022) also confirm the signaling effect of trade credit on bank loans.

To ensure that our findings are not solely influenced by concurrent shifts in bank credit, we divide the sample into three distinct groups based on the association between

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changes in trade credit and bank credit.[21] When a firm experiences simultaneous increases or decreases in both trade credit and bank financing, we interpret this as a positive relationship, indicating a higher likelihood of complementarity between the two. Conversely, if a firm increases one form of financing while decreasing the other, we consider this a negative relationship, suggesting a greater likelihood of substitution between them. The third group comprises firms with missing observations or no changes in either trade credit or bank credit. Subsequently, we apply Eq. (1) separately to each of these subsamples.

The results are presented in Panel B, Table 9. If the increase in trade credit after the reform is driven by bank financing, we would expect the results to be concentrated in the subsamples with a positive relationship between trade credit and bank credit. On the contrary, the estimated coefficients on *Reform* remain positive in all cases and are statistically significant in all cases except for the third group. As such, the relation between trade credit and bank financing does not play a significant role in the relationship between the SCR and trade credit financing.

The Impact of Additional Trade Credit on Firm Performance

So far, we have found a significant increase in trade credit financing, particularly for young, small, private, and politically unconnected firms following the SCR. As discussed earlier, trade credit is a vital financing channel supporting firm growth in China (Allen et al., 2019) and plays a more significant role, particularly for small private firms compared to developed markets (Ge & Qiu, 2007; Kong et al., 2020; Lai et al., 2022). A growing body of finance literature suggests that trade credit plays a more complex role than simply providing short-term financing and liquidity. For example, financially unconstrained firms relied more on bank loans during the Global Financial Crisis, while financially constrained SMEs turned to trade credit (Carbó-Valverde et al., 2016). Ferrando and Mulier (2013) show that trade credit is a valuable tool for young and fast-growing firms to expand output and manage their growth rate. Greater access to trade credit financing can also provide firms with the capacity to engage in strategic corporate decisions such as innovation (Shahzad et al., 2021), capital investment (Ding et al., 2013), and driving environmental improvement (An et al., 2021).

In this section, we examine whether the additional trade credit associated with social credit reform can have a real impact on firms. Specifically, we examine the effect of the increase in trade credit on firm investment strategy and performance in the coming year by using a two-stage model (Kim et al., 2021). First, we convert the baseline regression model into the following DID model in the first stage. The fitted values of the dependent variables (*ΔTC_Pre*) estimated in this model reflect the impact of the reform on changes in corporate trade credit:

$$\Delta TC_{i,t} = \alpha_0 + \beta_1 Reform_{i,t} + \gamma \Delta Z_{i,t} + \delta_i + \eta_s + \varepsilon_{i,t} \quad (3)$$

Next, we use the following regression to test the impact of the change in firms' trade credit caused by the reform on the change in firm investment strategy and performance in the coming year:

$$\Delta Strategy/Performance_{i,t+1} = \alpha_0 + \beta_1 \Delta TC_Pre_{i,t} + \gamma Z_{i,t} + \delta_i + \eta_s + \varepsilon_{i,t} \quad (4)$$

The dependent variable is either the change in firm investment strategy, financial performance, or environmental performance. We use R&D investment (*R&D*) and capital expenditures (*Capex*) to proxy for corporate investment strategy, and ROE and Tobin's Q to proxy for firm performance. The environment performance covers five areas including the energy saving score (*EneSav*), types of pollution score (*TP*), investment in environment protection score (*IEP*), the accreditation of the environment management system score (*AEMS*), and environmental consciousness score (*EnvCon*). Firm performance data come from the *CSMAR* database, while environmental data are from *Hexun.com*. The key variable of interest is the fitted value estimated by the first-stage regression model.

Table 10 presents the regression results for Eq. (4). Panel A reports the results of the real effect of the additional trade credit on changes in firm investment strategy. Our key variable of interest is the *ΔTC_Pre*. The coefficient β_1 is positive and significant, indicating that the reform's expansion of trade credit boosts firms' R&D investment and capital investment. Panel B shows the impact of the reform on firm performance through the channel of trade credit. The coefficients of *ΔTC_Pre* are positive and significant, indicating that the reform's increase in trade credit has a positive impact on firm performance as measured by changes in ROE (*ΔROE*) and Tobin's Q (*ΔTobin*). Panel C shows the impact of changes in firms' trade credit caused by the reform on changes in firm environmental performance. The coefficients of *ΔTC_Pre* are positive and significant in all columns, indicating that the reform's increase in trade credit leads to firms investing more in environmental initiatives. Overall, our findings suggest that the reform promotes firm investment strategy and performance, by improving firms' access to trade credit.

[21] Ideally, we would evaluate the bank credit channel by having staggered shocks to firms' access to bank credit similar to the identification strategy of Shenoy & Williams (2017). However, in the absence of appropriate shocks, we adopt an alternative setting.

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Table 10 Real effects in response to the recognition of the SCR

Dependent variable =	⊿*R&D* (1)	⊿*Capex* (2)
Panel A. The real effect on corporate strategy		
⊿*TC_Pre*	0.241***	0.825*
	(10.172)	(1.780)
Controls	Yes	Yes
Firm	Yes	Yes
Year	Yes	Yes
Adj. R^2	0.067	0.066
N	20,101	19,258
Dependent variable =	⊿*ROE* (1)	⊿*Tobin* (2)
Panel B. The real effect on firm performance		
⊿*TC_Pre*	2.345***	5.790*
	(2.973)	(1.844)
Controls	Yes	Yes
Firm	Yes	Yes
Year	Yes	Yes
Adj. R^2	0.042	0.247
N	20,101	19,071

Dependent variable =	⊿*EneSav* (1)	⊿*TP* (2)	⊿*IEP* (3)	⊿*AEMS* (4)	⊿*EnvCon* (5)
Panel C. The real effect on carbon reduction performance					
⊿*TC_Pre*	2.625***	2.648***	1.748***	0.212***	8.921***
	(13.633)	(13.845)	(11.608)	(3.011)	(13.662)
Controls	Yes	Yes	Yes	Yes	Yes
Firm	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes
Adj. R^2	0.081	0.083	0.087	0.018	0.094
N	18,583	18,583	18,583	18,583	18,583

This table presents coefficients from regressions of how the change of trade credit in response to the reform affects firm strategy and performance
Panel A reports the real effect of corporate strategy
The dependent variables are the changes in firm R&D investment and capital expenditure
Panel B reports the real effect of corporate performance, which is measured by Tobin's Q and ROE, respectively
Panel C reports the real effect of carbon reduction performance, which is measured by energy savings (*EneSav*), types of pollution (*TP*), investment in environmental protection (*IEP*), the accreditation of the environment management system (*AEMS*), and environmental consciousness (*EnvCon*)
All other variables are defined in Appendix 3
All regressions control for firm and year fixed effects
T-statistics are based on standard errors adjusted for heteroscedasticity and clustered at the firm level
*, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively

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Conclusion

In this study, we use the rollout of China's social credit reform as a natural experiment to examine the coevolution of ethical business credit practices, credit decisions of suppliers and customers, and institutional transitions. Based on a sample of Chinese A-share listed firms from 2011 to 2019, we find that the reform improves firms' access to trade credit through improved ethical credit behavior, including reduced default risks enhanced information quality, and improved legal protection. Evidence of suppliers' credit decisions coevolving with customers' ethical credit behaviors through a positive feedback loop is also presented. Further, we observe this effect to be particularly pronounced for firms facing traditional difficulties in securing trade credit, such as newer and smaller enterprises, private firms, and those lacking political affiliations. The effect of the reform is also more significant in pilot firms with geographically distant suppliers. Lastly, our analysis suggests that the increased trade credit financing resulting from the reform contributes to more R&D and fixed investments and better firm environmental and financial performance.

This research provides empirical evidence that in China's emerging market, characterized by significant credit rationing and severe information asymmetry, social credit plays a crucial role in facilitating firms' access to trade credit. Our findings contribute to a deeper understanding of the factors influencing trade credit dynamics, demonstrating how an information-sharing platform, integrated with 'social monitoring,' shapes economic activity by influencing firms' ability to both extend and obtain informal financing. These findings offer valuable insights into the remarkable growth of the Chinese economy, particularly considering the limitations of well-developed formal institutions and the varying levels of access to formal financing channels for small and private enterprises.

Our research has significant implications for policymakers, especially in developing markets, as it provides compelling evidence that social credit can shift the boundaries of business ethics by rewarding ethical credit behavior while penalizing unethical practices. Future research can focus on directly examining the impact of social credit evolution on corporate behavior, contingent upon the standardization and availability of data on blacklists and whitelists. Furthermore, a comprehensive understanding requires further investigation into the broader societal and ethical implications, including potential concerns such as privacy issues and potential disparities in evaluation metrics.

Appendix

Appendix 1. Pilot City Distribution

Panel A shows the distribution of pilot cities (or districts) for the SCR in China between 2015 and 2016. Panel B presents the annual distribution of companies located in the pilot areas of the SCR. Panel C visualizes the geographic locations of pilot cities (or districts) and sample firms.

Panel A. City (or district) distribution

Region	Date	Region	Date
Shenyang	2015	Liaoyang	2016
Qingdao	2015	Suifenhe	2016
Nanjing	2015	Pudong New District, Shanghai	2016
Wuxi	2015	Suzhou	2016
Suqian	2015	Taizhou	2016
Hangzhou	2015	Anqing	2016
Wenzhou	2015	Huaibei	2016
Yiwu	2015	Fuzhou	2016
Hefei	2015	Xiamen	2016
Wuhu	2015	Putian	2016
Chengdu	2015	Weifang	2016
Haidian District, Beijing	2016	Weihai	2016
Hohhot	2016	Dezhou	2016
Wuhai	2016	Rongcheng	2016
Dalian	2016	Zhengzhou	2016
Anshan	2016	Nanyang	2016
Wuhan	2016	Shenzhen	2016
Xianning	2016	Zhuhai	2016
Yichang	2016	Shantou	2016
Huangshi	2016	Huizhou	2016
Guangzhou	2016	Luzhou	2016
Jiading District, Shanghai	2016		

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Panel B. Descriptive statistic of sample firms

Year	*Reform* = 1		*Reform* = 0		Total	
	Obs N	Freq %	Obs N	Freq %	Obs N	Freq %
2011	0	0	1864	100	1864	8.767
2012	0	0	2142	100	2142	10.074
2013	0	0	2306	100	2306	10.846
2014	0	0	2359	100	2359	11.095
2015	301	12.361	2134	87.639	2435	11.452
2016	1141	44.088	1447	55.912	2588	12.172
2017	1123	44.074	1425	55.926	2548	11.984
2018	1113	44.062	1413	55.938	2526	11.880
2019	1093	43.825	1401	56.175	2494	11.730
Total	4771	22.439	16,491	77.561	21,262	100

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Panel C. Geographical plot



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Appendix 2. Local Credit China Website

This appendix showcases examples from two local Credit China websites. Panel A demonstrates a local social credit website, featuring company information and black/white lists for businesses in Guangzhou, Guangdong Province. Panel B provides specific details about a blacklisted firm located in Gongyi City, Henan Province.

Panel A. Local social credit website



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Panel B. Details of black list



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Appendix 3. Variable Definitions

Variable	Definition
TC_{t+1}	The net amount of accounts payable and accounts receivable scaled by total assets in year$t + 1$
$TC2_{t+1}$	Total accounts payable divided by total assets in year $t + 1$
$TC3_{t+1}$	Total accounts payable divided by total liabilities in year $t + 1$
Reform	A dummy variable equals 1 when the company is located in a region chosen for piloting SCR, including the years following the initial rollout, and 0 otherwise
Lev	Total liabilities divided by total assets
Size	Natural logarithm of total assets
Cash	The ratio of cash divided by total assets
ROA	Net profit divided by total assets
PPE	The ratio of the value of the net property, plants, and equipment divided by the value of total assets
Growth	The growth rate of sales
SOE	The dummy variable equals 1 if the firm is a state-owned enterprise, and 0 otherwise
Age	Natural logarithm of years since a firm's founding
R&D	Research and development expenses, scaled by sales
Board	Natural logarithm of the total number of directors
CFO	The ratio of cash flow from operating activities divided by total assets
Indep	The percentage of independent directors on the board
PC	A dummy variable that equals 1 if the CEO or Chairman of a firm is a former government official, a member of the National People's Congress (NPC), or a member of the Chinese People's Political Consultative Conference (CPPCC), and 0 otherwise
Default	The natural logarithm of one plus the number of a company's managerial default events such as failure to meet targets or goals, or neglecting legal or ethical responsibilities
DD	Distance to default calculated using a Merton Model (Bharath & Shumway, 2008), measuring how many standard deviations the current market value of a company's assets is away from a default point
Z-score	The Altman's Z-score is calculated as 6.56*(working capital/total assets) + 3.26*(retained earnings/total assets) + 6.72*(earnings before interests and taxes/total assets) + 1.05*(market value of equity/market value of total liabilities)
EQ	Earning quality as measured by discretionary accruals estimated using the Modified Jones (1991) model
IN	The negative of stock price synchronicity as measured in Morck (2000)
Attention	Natural logarithm of the number of analysts concerned
Market	An index that describes the development of the local marketization process from Fan et al. (2011)
Lawyer	The lawyer-to-population ratio, measured as the proportion of lawyers to the total population within a prefecture city
IPI	An index that describes the development of the local legal environment from Fan et al. (2011)
ROE	Net income divided by total equity
Tobin	Ratio of market value to total assets
TOP1	Shareholding percentage of the largest shareholder
Instl	The ratio of institutional holdings on outstanding shares
Dual	The dummy variable equals 1 if a firm's CEO is also the chairman of the board and 0 otherwise

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Appendix 4. Policy Effectiveness

This table represents the result of regressions that examine the effect of the implementation of SCR on city credit scores. The dependent variable *CEI* is derived from the "Blue Book of China City Business Credit Environment Index," indicating the credit environment of a city. *Reform* is a dummy variable that equals 1 if firms are registered at pilot cities during and after the event year, and 0 otherwise. Column (1) shows the baseline results. Column (2) reports the results of the parallel trend analysis, where *Reform* is replaced with several dummy variables that trace out the yearly effects of SCR. Other variables are defined in Appendix 3. T-statistics are based on standard errors adjusted for heteroscedasticity and clustered at the firm level. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

Dependent variable =	*CEI*	*CEI*
	(1)	(2)
Reform	2.117***	
	(21.397)	
Before3		0.145
		(1.269)
Before2		0.043
		(0.301)
Current		1.607***
		(13.270)
After1		1.327***
		(35.613)
After2		0.492***
		(3.019)
After3		1.882***
		(13.694)
Controls	Yes	Yes
Firm	Yes	Yes
Year	Yes	Yes
Adj. R^2	0.250	0.287
N	11,303	11,303

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Appendix 5. Univariate Analysis of the Matched Sample

This table shows the effectiveness of the propensity score matches. T-statistics are in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively. Variables are defined in Appendix 3.

		Treated	Control	Mean-Diff	T
		(1)	(2)	(3)	(4)
Lev	U	0.442	0.435	0.007*	1.83
	M	0.442	0.443	− 0.001	− 0.30
Size	U	22.455	22.172	0.283***	13.51
	M	22.455	22.432	0.023	0.87
Cash	U	0.170	0.179	− 0.009***	− 4.44
	M	0.170	0.168	0.002	0.54
ROA	U	0.029	0.034	− 0.005***	− 4.12
	M	0.029	0.030	− 0.001	− 0.17
PPE	U	0.182	0.228	− 0.046***	− 17.19
	M	0.182	0.182	0.000	0.00
Growth	U	0.186	0.175	0.011	1.56
	M	0.186	0.192	− 0.006	− 0.58
SOE	U	0.323	0.412	− 0.089***	− 11.09
	M	0.323	0.317	0.006	0.66
Age	U	2.341	2.209	0.132***	11.02
	M	2.341	2.351	− 0.010	− 0.74
R&D	U	0.020	0.015	0.005***	18.85
	M	0.020	0.020	0.000	0.50
Board	U	2.105	2.146	− 0.041***	− 12.81
	M	2.105	2.101	0.004	0.89
CFO	U	0.044	0.041	0.003***	2.95
	M	0.044	0.045	− 0.001	− 0.77
Indep	U	0.382	0.380	0.002*	1.65
	M	0.382	0.384	− 0.002	− 1.28

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Appendix 6. Predictive Analysis

This table examines whether a city's macroeconomic characteristics predict its inclusion as a pilot city. The dependent variable is *Reform*, a dummy that equals 1 for cities included as pilot cities in the post-reform years, and 0 for the pre-reform years, or for cities not included as pilot cities. *County_average_of_TC* is the average level of trade credit in county-level cities. *GDP_Growth* is the GDP growth rate of county-level cities. *Log_GDP_per_capita* is the natural log of GDP per capita. *Gov_expenditure* denotes the ratio of government expenditure divided by local GDP. *Fixed_asset_investment* is the ratio of fixed asset investment divided by local GDP. *GDP2/GDP3* is the ratio of secondary sector GDP to tertiary sector GDP in the city. *Edu* is the natural logarithm of the number of college students in the city. *Population* is the natural logarithm of the number of residents in the city. Cities initially included in the pilot program were removed from the sample after their participation. The sample period is from 2011 to 2019. All variable definitions are presented in Appendix 3. All continuous variables are winsorized at the 1st and 99th percentiles. T-statistics are based on standard errors adjusted for heteroscedasticity and clustered at the city level. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

Dependent variable =	*Reform*	*Reform*
	(1)	(2)
County_average_of_TC	− 0.100	0.110
	(− 1.261)	(0.622)
GDP_Growth	0.234	0.494
	(1.362)	(0.707)
Log_GDP_per_capita	0.063***	− 0.084
	(5.398)	(− 0.770)
Government_expenditure	0.131	− 0.271
	(1.619)	(− 0.785)
Fixed_asset_investment	− 0.052**	− 0.169
	(− 2.461)	(− 0.838)
GDP2/GDP3	− 0.001	0.059
	(− 0.155)	(0.727)
Edu	− 0.014	− 0.084
	(− 0.780)	(− 0.788)
Population	0.045**	0.099
	(2.416)	(1.121)
Constant	− 0.901***	0.717
	(− 5.353)	(0.588)
City	No	Yes
Year	No	Yes
Adj. R^2	0.139	0.185
N	6376	6376

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Appendix 7. Other Robustness Tests

This table presents other robustness results. Panel A controls for different fixed effects and applies alternative measures. Column (1) reports the results with additional city year-level variables. *Log_GDP_per_capita* is the natural log of GDP per capita. *Population* is the natural logarithm of the number of residents in the city. *Gov_expenditure* denotes the ratio of government expenditure divided by local GDP. Column (2) reports the result of the cluster robustness standard errors at the city level. Columns (3)-(4) report the results with industry $\times$ year fixed effect and city $\times$ year fixed effect , respectively. Columns (5)-(6) report the results using two alternative measures of trade credit (*TC2* and *TC3*). Panel B controls for contemporaneous shocks. To exclude the impact of the 4G network constructions, column (1) takes into account the effect by adding the 4G as the control variable. Column (2) further controls the staggered rollout of the establishment of Circuit Courts (*Court*). Column (3) reports the introduction of interest rate deregulation (*HP*). To exclude the impact of the deleveraging policy, column (4) takes into account the effect by adding the *Deleveraging* as the control variable, which takes the value of one for SOEs after 2015, and zero otherwise. Column (5) takes into account the effect by adding the *NECIPS* as the control variable, which equals one for firms with tangible asset ratios below the industry median at the end of 2013, in the period after 2014, and zero otherwise. Column (6) controls for the effect of digital social credit platforms. *Platform* is a dummy variable that equals one for cities that voluntarily adopt a digital social credit platform (She et al., 2024). Panel C presents the results of the Oster (2019) tests. The identified set includes the value of β in the controlled model (upper bound) and the value of β calculated for $R_{max} = 1.3R^2$, under the assumption that selection on observable and unobservable is proportional. Controls are the same as those in Table 2, and all variable definitions are shown in Appendix 3. T-statistics are based on standard errors adjusted for heteroscedasticity and clustered at the firm level except for column (2) of panel A, which is clustered at the city level. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

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Panel A. Applying different fixed effects and alternative measures

	Controlling for city-level variable	Cluster at City	Industry × year	City × year	Alternative measures	
Dependent variable =	TC_{t+1}	TC_{t+1}	TC_{t+1}	TC_{t+1}	$TC2_{t+1}$	$TC3_{t+1}$
	(1)	(2)	(3)	(4)	(5)	(6)
Reform	0.006***	0.006**	0.007***	0.006**	0.036***	0.005***
	(2.933)	(2.399)	(3.175)	(2.341)	(4.160)	(2.628)
Log_GDP_per_capita	0.009					
	(1.446)					
Population	- 0.006					
	(-0.873)					
Gov_expenditure	0.015					
	(0.426)					
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Firm	Yes	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes	Yes
Industry × Year	No	No	Yes	No	No	No
City × Year	No	No	No	Yes	No	No
Adj. R^2	0.774	0.773	0.777	0.776	0.576	0.770
N	20,810	21,262	21,262	21,262	21,262	21,262

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Panel B. Controlling for contemporaneous policies

Dependent variable =	TC_{t+1}	TC_{t+1}	TC_{t+1}	TC_{t+1}	TC_{t+1}	TC_{t+1}
	(1)	(2)	(3)	(4)	(5)	(6)
Reform	0.006***	0.006***	0.006***	0.006***	0.006***	0.008***
	(2.616)	(2.848)	(2.828)	(2.917)	(2.598)	(2.621)
4G	0.004*					
	(1.773)					
Court		0.001				
		(0.816)				
HP			- 0.001			
			(-0.705)			
Deleveraging				0.004		
				(1.528)		
NECIPS					0.013***	
					(5.810)	
Platform						0.002
						(0.620)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Firm	Yes	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R^2	0.773	0.773	0.773	0.773	0.773	0.774
N	21,262	21,262	21,262	21,262	21,262	21,262

Panel C. Controlling for omitted variables and regression bias

Dependent variable = TC_{t+1}		
Standard	Estimated value	Omitted variables bias
$\beta^*(R_{max}, \delta) \in [0.002,0.010]$	$\beta^*(R_{max}, \delta) = 0.006$	Unlikely
$\delta > 1$	$\delta = 5.46$	Unlikely

Declarations

Conflict of interest The authors have no conflict of interest to declare that are relevant to the content of this article. All persons who meet authorship criteria are listed as authors, and all authors certify that they have participated sufficiently in the work to take public responsibility for the content, including participation in the concept, design, analysis, writing, or revision of the manuscript. Furthermore, each author certifies that this material or similar material has not been and will not be submitted to or published in any other publication before its appearance in the *Journal of Business Ethics*.

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